



RegSphere AI Credit Scoring Governance Tool: A jurisdiction-aware RegTech prototype for AI lending governance across US and EU regulatory expectations

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Why This Product Is Right for RegSphere

RegSphere AI Governance Solutions focuses on AI governance and cross-jurisdictional compliance monitoring for financial institutions.

This product fits the company because it turns RegSphere's core values into operational controls:

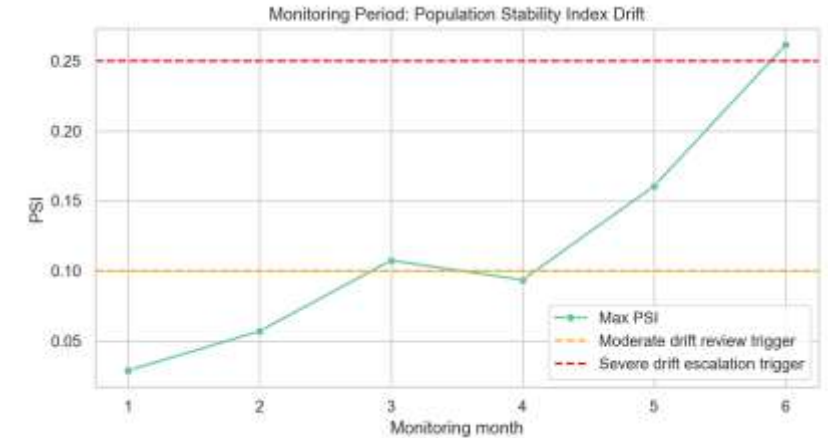
- Fairness → approval disparity, FPR gap, FNR gap
- Transparency → feature importance and rejection reasons
- Accountability → compliance report and governance documentation
- Regulatory adaptability → US/EU jurisdiction rule engine
- Responsible automation → alerts support human review, not replacement



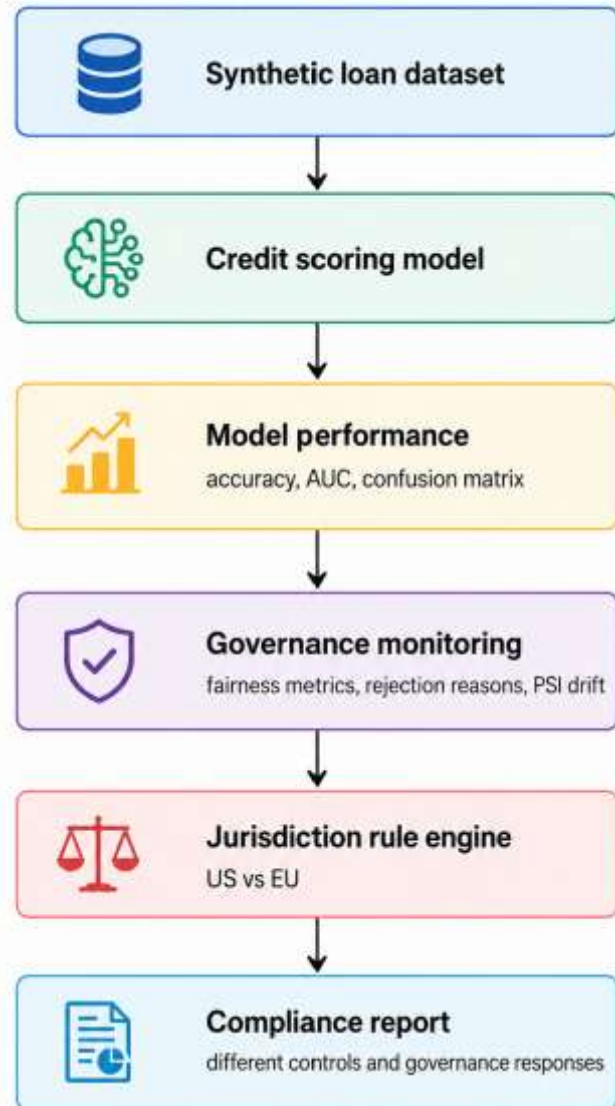
What This Tool Solves That a Memo or Spreadsheet Cannot

AI credit scoring risk is dynamic. A static memo or spreadsheet cannot continuously monitor whether model behavior changes after deployment.

Traditional Static Compliance Tools	RegSphere Prototype
One-time documentation	Generates synthetic lending data
Manual updates	Trains a credit scoring model
Difficult to track drift	Monitors fairness metrics
Hard to compare jurisdictions	Detects model drift using PSI
Weak escalation trail	Produces jurisdiction-specific compliance outputs
	Exports tables and charts for review



How the Working Prototype Operates



Core output files:

Task3_demo.ipynb

synthetic_loan_data.csv

monitoring_results.csv

jurisdiction_compliance_report.csv

model_card_governance_stub.md

Regulatory Divergence: What It Handles and How

Category	United States	European Union
Main Regulation	ECOA, Regulation B, CFPB guidance	EU AI Act, Regulation (EU) 2024/1689
Core Objective	Fair lending and anti-discrimination	High-risk AI governance
Key Requirements	Specific and accurate adverse action explanations	Human oversight and transparency
Monitoring Focus	Disparate impact / fairness review	Technical documentation and monitoring logs
Governance Response	Fair lending review	High-risk AI escalation

How the tool handles the divergence:

1. Stores US and EU jurisdiction-specific governance parameters in a rule engine.
2. Applies the active jurisdiction configuration to the same model metrics.
3. Converts metric breaches into different governance responses.
4. Exports the result in jurisdiction_compliance_report.csv for human review.

Example:

- Latest monitoring PSI = 0.2614
- US threshold = 0.30 → drift control not triggered
- EU threshold = 0.10 → drift breach triggered

Important caveat: Thresholds are internal governance thresholds, not statutory thresholds.

What the Tool Does Not Do

The tool supports governance review. It does not outsource judgement.

- It does not make final credit approval decisions.
- It does not prove legal discrimination.
- It does not issue final adverse action notices.
- It does not replace legal, compliance, or model risk professionals.
- It does not use real customer data in this prototype.
- It does not claim that its thresholds are statutory legal thresholds.

Why this boundary matters:

Fairness, discrimination, and legal compliance require context-specific human judgement. The prototype is designed to surface risks earlier and more consistently, while preserving human accountability.